Chapter 20 exercises • Influential operating systems

OS20-E01 • Spool bottleneck

Three spooled jobs each need CPU 2 then printer 4;CPU/printer overlap, one printer.

1. List all intervals.
2. Compare total/utilization with serial execution.
3. Distinguish spooling from multiprogramming.

OS20-E02 • THE and Mach boundaries

Compare THE software layers with a Mach3 user server.

1. What responsibilities do both decompose?
2. Which specified path entails address-space crossing?
3. State conditions for layer-contract induction.

OS20-E03 • Atlas branches and faults

FramesA(0,2), B(0,5), C(0,3), then a separate setA(2,4), B(1,6), C(0,8), where pairs are(t, T).

1. Select victims/branches.
2. For512-word pages, VA3597, page 7→frame 2, findPA.
3. Compare invariants with modern demand paging.

OS20-E04 • XDS relocation/sharing

Virtual16 Kiwords, physical64 Kiwords, pages 2 Kiwords, P page 3→frame 11. Two processes each use eight pages, including two shareable code pages.

1. Count pages/frames and map VA6161.
2. Count frames after sharing.
3. Does a page table prove demand paging?

OS20-E05 • Two-buffer conservation

Poolb0/b1;C1/C2 send requests, S replies to C1 usingb0, not yet consumed.

1. List free, queued, reply sets.
2. Can C3 claimb0 now? After C1 returns it?
3. Do atomic primitives prove deadlock freedom?

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Chapter 20 exercises • Influential operating systems (continued)

OS20-E06 • Exponential feedback

q_i=2^i, start0, burst11, demote on full-slice exhaustion before completion, no competition/promotion. Swap cost.5 per dispatch.

1. Give allowed/used/remaining per dispatch.
2. Compare total with RR q1.
3. Derive cumulative service and limits of the trace model.

OS20-E07 • Multics segments/pages

OriginalGE645 has18-bit segment and18-bit word offset,1024-word pages. Segment5 offset 2500 maps page 2→frame 7.

1. Split offset and calculatePA.
2. Why is a ring not a THE layer?
3. Why does resolved filename not authorize writing?

OS20-E08 • MFT/MVT and base-limit

Fixed four64 KiB regions, jobs50/50/70. Variable holes50/30/40. Toy base256, limit768, physical RAM1024.

1. Identify fragmentation.
2. Evaluate user addresses0,767,768,−1.
3. Distinguish CP/CMS roles.

OS20-E09 • Specified Mach3 copying

C sends32 B, S replies4096 B, twice-copied through kernel each direction, separate address spaces, no shared-page optimization.

1. Count mode/address-space changes and copy bytes.
2. What does server-crash isolation guarantee or leave unresolved?
3. Why not apply it to Mach2.x/every derivative?

OS20-E10 • Names versus mechanisms

A GUI has two windows;A loops in user code without yielding, B is ready. Elsewhere command completion succeeds but target lacks write authority.

Chapter 20 exercises • Influential operating systems (continued)

1. Compare cooperative/preemptive claims.
2. Does completion authorize writing?
3. Do high-level language/small kernel alone prove correctness?

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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